

Formula Sheet Part 1: Foundations, Co-Movement, and Portfolios

This is a two-column topic split from `All_Formulas_From_Chatnotes.md`.

Conventions

$$t, T, h, k, n, p, q, N$$

Variables: t is a time index; T is sample size or the final observed time; h is a forecast horizon; k is a lag or number of model parameters depending on context; n is a return horizon; p, q are lag orders; N is the number of assets.

$$F_t$$

Variables: F_t is the information set available at time t .

Returns may be written as decimals or percentages. Check the scale before applying VaR or standard deviation formulas.

Prices, Returns, and VaR

Simple return

$$R_t = \frac{P_t - P_{t-1}}{P_{t-1}} = \frac{P_t}{P_{t-1}} - 1$$

Variables: R_t is the simple return from $t-1$ to t ; P_t is the price at time t ; P_{t-1} is the previous price.

Gross return

$$1 + R_t = \frac{P_t}{P_{t-1}}$$

Variables: $1 + R_t$ is the gross return, the multiplier applied to wealth.

Return with dividends

$$R_t = \frac{P_t + D_t - P_{t-1}}{P_{t-1}}$$

Variables: D_t is the dividend paid during period t .

Log return

$$r_t = \log(1 + R_t) = \log(P_t) - \log(P_{t-1}) = \log\left(\frac{P_t}{P_{t-1}}\right)$$

Variables: r_t is the log return; $\log$ is the natural logarithm.

Small-return approximation

$$\log(1 + R_t) \approx R_t$$

Variables: R_t is the simple return; the approximation works best when R_t is small.

Multi-period gross return

$$1 + R_t(k) = \prod_{j=0}^{k-1} (1 + R_{t-j})$$

Variables: $R_t(k)$ is the k -period simple return ending at t ; j indexes the returns being multiplied.

Multi-period simple return

$$R_t(k) = \prod_{j=0}^{k-1} (1 + R_{t-j}) - 1$$

Variables: $R_t(k)$ is the total simple return over k periods.

Equity curve for one dollar

$$V_t(k) = \prod_{j=0}^{k-1} (1 + R_{t-j})$$

Variables: $V_t(k)$ is the value at time t of one dollar invested over k periods.

Wealth growth

$$V_T = V_0 \prod_{t=1}^T (1 + R_t)$$

Variables: V_0 is starting wealth; V_T is ending wealth after T periods.

Multi-period log return

$$r_t(k) = \sum_{j=0}^{k-1} r_{t-j} = r_t + r_{t-1} + \dots + r_{t-k+1}$$

Variables: $r_t(k)$ is the k -period log return.

Total log return between two prices

$$r(1, T) = \sum_{t=1}^T r_t = \log\left(\frac{P_T}{P_0}\right)$$

Variables: P_0 is the starting price; P_T is the ending price.

Index log return

$$r_t = \log(Index_t) - \log(Index_{t-1})$$

Variables: $Index_t$ is the index level at time t .

Excess returns

$$\text{asset excess return} = \text{asset return} - R_f$$

$$\text{market excess return} = \text{market return} - R_f$$

Variables: R_f is the risk-free return.

Market beta interpretation

$$\beta > 1 \Rightarrow \text{aggressive, more sensitive than the market}$$

$$\beta = 1 \Rightarrow \text{tracks the market}$$

$$0 < \beta < 1 \Rightarrow \text{defensive, less sensitive than the market}$$

$$\beta = 0 \Rightarrow \text{unrelated to market movements}$$

Variables: β is market sensitivity.

Market capitalisation

$$\text{market capitalisation} = \text{share price} \times \text{number of shares outstanding}$$

Variables: market capitalisation is firm size measured using market value.

Book-to-market ratio

$$\text{BtM} = \frac{\text{book value of equity}}{\text{market value of equity}}$$

Variables: BtM is the book-to-market ratio.

Normal return model

$$R \sim \mathcal{N}(\mu, \sigma^2)$$

Variables: R is a return; μ is mean return; σ^2 is variance; σ is standard deviation.

Normal quantile

$$q_\alpha = \mu + \sigma z_\alpha$$

Variables: q_α is the lower-tail return quantile; α is the tail probability; z_α is the standard normal cutoff.

Common normal cutoffs

$$z_{0.05} = -1.645$$

$$z_{0.025} = -1.96$$

Variables: $z_{0.05}$ is the 5 percent lower-tail cutoff; $z_{0.025}$ is the 2.5 percent lower-tail cutoff.

One-asset VaR

$$\text{VaR}_\alpha = |W_0 q_\alpha|$$

Variables: VaR_α is Value at Risk at tail probability α ; W_0 is the dollar value invested.

Zero-mean 5 percent normal VaR

$$q_{0.05} = -1.645\sigma$$

$$\text{VaR}_{0.05} = |W_0(-1.645\sigma)|$$

Variables: $q_{0.05}$ is the 5 percent return cutoff; σ is return volatility.

Random Variables and Distribution Properties

Probability under a density

$$\Pr(a \leq X \leq b) = \text{area under } f(x) \text{ from } a \text{ to } b$$

Variables: X is a random variable; a, b are cutoffs; $f(x)$ is the probability density.

Normal distribution notation

$$X \sim \mathcal{N}(\mu, \sigma^2)$$

Variables: X is a random variable; μ is the mean; σ^2 is the variance.

Expected value

$$\mathbb{E}[X] = \mu$$

Variables: $\mathbb{E}[\cdot]$ means expected value.

State-contingent expected return

$$\mathbb{E}(R) = \sum_s R_s \Pr(S = s)$$

Variables: R_s is the return in state s ; S is the state variable.

Expected value linearity

$$\mathbb{E}(a) = a$$

$$\mathbb{E}(aX) = a\mathbb{E}(X)$$

$$\mathbb{E}(a + X) = a + \mathbb{E}(X)$$

$$\mathbb{E}(X + Y) = \mathbb{E}(X) + \mathbb{E}(Y)$$

Variables: a is a constant; X, Y are random variables.

Variance

$$\text{Var}(X) = \mathbb{E}[(X - \mu)^2]$$

Variables: $\text{Var}(X)$ is the variance of X .

Standard deviation

$$\text{sd}(X) = \sqrt{\text{Var}(X)}$$

Variables: $\text{sd}(X)$ is the standard deviation of X .

Variance rules

$$\text{Var}(a) = 0$$

$$\text{Var}(aX) = a^2 \text{Var}(X)$$

$$\text{Var}(a + X) = \text{Var}(X)$$

Variables: a is a constant.

Variance of a sum

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2 \text{Cov}(X, Y)$$

Variables: $\text{Cov}(X, Y)$ is covariance between X and Y .

Standardising a normal variable

$$Z = \frac{X - \mu}{\sigma} \sim \mathcal{N}(0, 1)$$

Variables: Z is a standard normal variable; σ is the standard deviation of X .

Probability conversion after standardising

$$\Pr(X > a) = \Pr\left(\frac{X - \mu}{\sigma} > \frac{a - \mu}{\sigma}\right) = \Pr(Z > z)$$

Variables: $z = (a - \mu)/\sigma$ is the standardised cutoff.

Lower-tail normal probability

$$\Pr(r_t < x) = \Pr\left(Z < \frac{x - \mu}{\sigma}\right)$$

Variables: r_t is a return; x is a cutoff.

Return scale convention

0.01 = 1 percent, if returns are decimals

1 = 1 percent, if returns are stored as percentage points

Variables: the same economic return can be represented on different numeric scales.

Sample mean

$$\bar{r} = \frac{1}{T} \sum_{t=1}^T r_t$$

Variables: $\bar{r}$ is the sample average return.

Sample variance and standard deviation

$$s^2 = \frac{1}{T-1} \sum_{t=1}^T (r_t - \bar{r})^2, \quad s = \sqrt{s^2}$$

Variables: s^2 is sample variance; s is sample standard deviation.

Skewness and kurtosis

$$\text{skewness} = \frac{\mathbb{E}[(r_t - \mu)^3]}{\sigma^3}, \quad \text{kurtosis} = \frac{\mathbb{E}[(r_t - \mu)^4]}{\sigma^4}$$

Variables: skewness measures asymmetry; kurtosis measures tail thickness.

Co-Movement, Conditional Moments, and Factors

Joint probability

$$\Pr(X = x, Y = y)$$

Variables: X, Y are random variables; x, y are outcomes.

Marginal probability

$$\Pr(X = x) = \sum_y \Pr(X = x, Y = y)$$

Variables: the sum is over all possible values of Y .

Conditional probability

$$\Pr(X = x \mid Y = y) = \frac{\Pr(X = x, Y = y)}{\Pr(Y = y)}$$

Variables: $\Pr(X = x \mid Y = y)$ is the probability of $X = x$ given $Y = y$.

Independence

$$\Pr(X = x \mid Y = y) = \Pr(X = x)$$

Variables: independence means knowing Y does not change the probability distribution of X .

Conditional expectation

$$\mathbb{E}[X \mid Y]$$

Variables: $\mathbb{E}[X \mid Y]$ is the expected value of X conditional on Y .

Law of iterated expectations

$$\mathbb{E}[X] = \mathbb{E}[\mathbb{E}[X \mid Y]]$$

Variables: Y is the conditioning variable.

Conditional variance

$$\text{Var}(Y \mid X) = \mathbb{E}[(Y - \mathbb{E}[Y \mid X])^2 \mid X]$$

Variables: $\text{Var}(Y \mid X)$ is remaining uncertainty about Y after conditioning on X .

Law of total variance

$$\text{Var}(Y) = \mathbb{E}[\text{Var}(Y \mid X)] + \text{Var}(\mathbb{E}[Y \mid X])$$

Variables: the first term is average conditional variance; the second is variance of conditional means.

Information reduces conditional variance

$$\text{Var}(Y \mid \text{information}) < \text{Var}(Y)$$

Variables: information is any conditioning set that improves prediction.

Covariance

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])]$$

Variables: covariance measures linear co-movement.

State covariance

$$\text{Cov}(R_A, R_B) = \sum_s p_s (R_{A,s} - \mu_A)(R_{B,s} - \mu_B)$$

Variables: R_A, R_B are asset returns; p_s is the probability of state s ; μ_A, μ_B are expected returns.

Covariance from correlation

$$\text{Cov}(R_A, R_B) = \rho_{AB} \sigma_A \sigma_B$$

Variables: ρ_{AB} is the correlation between assets A and B; σ_A, σ_B are standard deviations.

Correlation

$$\text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\text{sd}(X) \text{sd}(Y)}$$

Variables: $\text{Corr}(X, Y)$ is the standardised covariance.

Correlation from covariance

$$\rho_{AB} = \frac{\text{Cov}(R_A, R_B)}{\sigma_A \sigma_B}$$

Variables: ρ_{AB} is the correlation coefficient.

Covariance sign interpretation

$\text{Cov}(X, Y) > 0 \Rightarrow X$ and Y tend to move together

$\text{Cov}(X, Y) < 0 \Rightarrow X$ and Y tend to move oppositely

$\text{Cov}(X, Y) = 0 \Rightarrow$ no linear co-movement

Variables: the sign of covariance gives direction of linear co-movement.

Group mean return

$$\bar{r}_g = \frac{1}{T} \sum_{t=1}^T r_{g,t}$$

Variables: $\bar{r}_g$ is the mean return for group g ; $r_{g,t}$ is group return at time t .

Generic factor long-short return

$$r_{\text{factor},t} = r_{\text{long},t} - r_{\text{short},t}$$

Variables: $r_{\text{long},t}$ is the return on the long leg; $r_{\text{short},t}$ is the return on the short leg.

SMB factor

$$SMB_t = R_{\text{small},t} - R_{\text{big},t}$$

Variables: SMB_t is Small Minus Big at time t .

HML factor

$$HML_t = R_{\text{high BtM},t} - R_{\text{low BtM},t}$$

Variables: HML_t is High Minus Low book-to-market at time t .

Fama-French two-by-three SMB construction

$$SMB = \text{average}(S/L, S/N, S/H) - \text{average}(B/L, B/N, B/H)$$

Variables: S means small; B means big; L, N, H mean low, neutral, and high book-to-market portfolios.

Fama-French two-by-three HML construction

$$HML = \text{average}(S/H, B/H) - \text{average}(S/L, B/L)$$

Variables: S/H is the small high-BtM portfolio; B/H is the big high-BtM portfolio; $S/L, B/L$ are low-BtM portfolios.

Portfolio Risk and Sharpe Ratios

Two-asset portfolio return

$$R_p = wR_1 + (1-w)R_2$$

Variables: R_p is portfolio return; w is the weight in asset 1; R_1, R_2 are asset returns.

Two-asset expected return

$$\mathbb{E}[R_p] = w\mathbb{E}[R_1] + (1-w)\mathbb{E}[R_2]$$

Variables: $\mathbb{E}[R_p]$ is expected portfolio return.

Two-asset expected return using means

$$\mu_p = w\mu_1 + (1-w)\mu_2$$

Variables: μ_p is portfolio mean return; μ_1, μ_2 are asset mean returns.

Two-asset portfolio variance

$$\text{Var}(R_p) = w^2\sigma_1^2 + (1-w)^2\sigma_2^2 + 2w(1-w)\sigma_{12}$$

Variables: σ_1^2, σ_2^2 are asset variances; σ_{12} is covariance.

Two-asset portfolio variance using explicit weights

$$\sigma_p^2 = w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1w_2\sigma_{12}$$

Variables: w_1, w_2 are portfolio weights; σ_p^2 is portfolio variance.

Covariance from correlation

$$\sigma_{12} = \rho_{12}\sigma_1\sigma_2$$

Variables: ρ_{12} is correlation between assets 1 and 2.

Sharpe ratio

$$SR = \frac{\text{mean return} - \text{risk-free rate}}{\text{standard deviation}}$$

Variables: SR is the Sharpe ratio.

Asset Sharpe ratio

$$SR_i = \frac{\mu_i - r_f}{\sigma_i}$$

Variables: μ_i is expected return on asset i ; r_f is the risk-free rate; σ_i is standard deviation.

Portfolio Sharpe ratio

$$SR_p = \frac{\mu_p - r_f}{\sigma_p}$$

Variables: SR_p is portfolio Sharpe ratio; σ_p is portfolio standard deviation.

Two-asset GMV weight

$$w_1^{GMV} = \frac{\sigma_2^2 - \sigma_{12}}{\sigma_1^2 + \sigma_2^2 - 2\sigma_{12}}$$

Variables: w_1^{GMV} is the global minimum variance weight in asset 1.

Second GMV weight

$$w_2^{GMV} = 1 - w_1^{GMV}$$

Variables: w_2^{GMV} is the GMV weight in asset 2.

Regression trick for GMV weights

$$r_{2,t} = \beta_0 + \beta_1(r_{1,t} - r_{1,t}) + e_t$$

Variables: $r_{1,t}, r_{2,t}$ are asset returns; β_0, β_1 are regression coefficients; e_t is the error.

$$\beta_1 = w_1, \quad w_2 = 1 - \beta_1$$

Variables: β_1 gives the weight in asset 1 under the regression trick.

Regression slope form used in practice

$$b = \frac{\text{Cov}(R_1, R_2)}{\text{Var}(R_2)}$$

Variables: b is a regression slope in the practice note.

Many-asset GMV weights

$$\mathbf{w}^{GMV} = \frac{\Sigma^{-1}\mathbf{1}}{\mathbf{1}'\Sigma^{-1}\mathbf{1}}$$

Variables: $\mathbf{w}^{GMV}$ is the GMV weight vector; Σ is the covariance matrix; $\mathbf{1}$ is a vector of ones.

Portfolio return from GMV weights

$$r_{p,t} = w_1r_{1,t} + (1 - w_1)r_{2,t}$$

Variables: $r_{p,t}$ is portfolio return at time t .